CEC 315 – Signals and Systems: Exam 3 Study Guide

Lectures 16–23 | Rogelio Gracia Otalvaro | Spring 2026

Exam format: 50 min, 100 pts. Part I: 10 MC ($4 \times 10 = 40$ pts). Part II: multi-part problems (60 pts). **Always state the ROC with every transform.**

The Laplace Transform (Lctrs 16–17)

Definition & the s-Plane

Bilateral Laplace Transform

$$X(s) = \int_{-\infty}^{+\infty} x(t)\, e^{-st}\, dt, \quad s = \sigma + j\omega$$

- $\sigma = 0 \Rightarrow X(j\omega)$ (the Fourier transform).
- LHP: $\sigma < 0$. RHP: $\sigma > 0$. $j\omega$-axis: FT lives here.

ROC Properties (Memorize)

Region of Convergence Rules

1. ROC = vertical strip (depends only on σ).
2. ROC never contains poles.
3. Finite-duration, abs. integrable $\Rightarrow$ ROC = all s.
4. **Right-sided** $\Rightarrow$ $\mathrm{Re}\{s\} > \max_i \mathrm{Re}\{p_i\}$.
5. **Left-sided** $\Rightarrow$ $\mathrm{Re}\{s\} < \min_i \mathrm{Re}\{p_i\}$.
6. **Two-sided** $\Rightarrow$ strip between consecutive poles.
7. $j\omega$-axis $\in$ ROC $\Rightarrow$ Fourier transform exists.

$\frac{1}{s+a}$ is ambiguous without the ROC

Could be $e^{-at}u(t)$ *or* $-e^{-at}u(-t)$.

Laplace Transform Pairs

$x(t)$	$X(s)$	ROC
$\delta(t)$	1	all s
$u(t)$	$1/s$	$\sigma > 0$
$e^{-at}u(t)$	$\dfrac{1}{s+a}$	$\sigma > -\mathrm{Re}\{a\}$
$-e^{-at}u(-t)$	$\dfrac{1}{s+a}$	$\sigma < -\mathrm{Re}\{a\}$
$t\, e^{-at}u(t)$	$\dfrac{1}{(s+a)^2}$	$\sigma > -\mathrm{Re}\{a\}$
$\dfrac{t^{n-1}}{(n-1)!}e^{-at}u(t)$	$\dfrac{1}{(s+a)^n}$	$\sigma > -\mathrm{Re}\{a\}$
$\delta(t-t_0)$	e^{-st_0}	all s
$e^{-at}\cos\omega_0 t\; u(t)$	$\dfrac{s+a}{(s+a)^2+\omega_0^2}$	$\sigma > -a$
$e^{-at}\sin\omega_0 t\; u(t)$	$\dfrac{\omega_0}{(s+a)^2+\omega_0^2}$	$\sigma > -a$

Laplace Properties

Property	Time	s-Domain
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$
Time shift	$x(t-t_0)$	$e^{-st_0}X(s)$
s-shift	$e^{s_0 t}x(t)$	$X(s-s_0)$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X(s/a)$
Convolution	$x_1 * x_2$	$X_1 X_2$
Differentiation	dx/dt	$sX(s)$
Diff. in s	$-tx(t)$	dX/ds

Inverse Laplace: Partial Fractions

1. Decompose $X(s)$ into $\frac{A_i}{s-p_i}$ terms (use **cover-up method**).
2. Invert each term with the table.
3. **ROC decides direction:** ROC right of pole $\to u(t)$; ROC left of pole $\to u(-t)$.

Three pole types:

- *Distinct real:* each gives $Ae^{p_i t}u(\cdot)$.
- *Repeated* (order n): need n terms $\frac{B_1}{s-p} + \frac{B_2}{(s-p)^2} + \cdots + \frac{B_n}{(s-p)^n}$
  $\frac{1}{(s+a)^k} \leftrightarrow \frac{t^{k-1}}{(k-1)!}e^{-at}u(t)$.
- *Complex conj.*: complete the square, use cos / sin pairs.

Initial-Value & Final-Value Theorems

IVT / FVT (Laplace)

$$x(0^+) = \lim_{s\to\infty} sX(s)$$
$$x(\infty) = \lim_{s\to 0} sX(s)$$

FVT valid **only if** all poles of $sX(s)$ have $\mathrm{Re}\{p\} < 0$.

CT System Analysis (Lctr 18)

Transfer Function $H(s)$

From an ODE $\sum a_k \frac{d^k y}{dt^k} = \sum b_k \frac{d^k x}{dt^k}$: replace $d^k/dt^k \to s^k$:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_M s^M + \cdots + b_0}{a_N s^N + \cdots + a_0}$$

Stability (CT, Causal)

Golden Rule – CT

Causal + BIBO stable $\iff$ **all poles in the open LHP** ($\mathrm{Re}\{p_i\} < 0\ \forall i$).

Pole location	Status
All $\mathrm{Re}\{p_i\} < 0$ (LHP)	Stable
Any $\mathrm{Re}\{p_i\} > 0$ (RHP)	Unstable
On $j\omega$-axis, none in RHP	Marginally stable

Block Diagrams

- **Series:** $H = H_1 \cdot H_2$ **Parallel:** $H = H_1 + H_2$
- **Negative feedback:** $Q = \dfrac{G}{1+GF}$ **Positive:** $Q = \dfrac{G}{1-GF}$

Unilateral Laplace Transform

Lower limit changes to 0^-; captures ICs:

Unilateral Differentiation Property

$$\frac{dy}{dt} \xrightarrow{\mathcal{L}_u} sY(s) - y(0^-)$$
$$\frac{d^2y}{dt^2} \xrightarrow{\mathcal{L}_u} s^2Y(s) - sy(0^-) - y'(0^-)$$

Total response = ZSR (input, zero ICs) + ZIR (ICs, zero input).

The z-Transform (Lctrs 19–20)

Definition

z-Transform

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n]\, z^{-n}, \quad z = re^{j\omega}$$

- $|z| = 1 \Rightarrow X(e^{j\omega})$ (the DTFT).
- Unit circle replaces the $j\omega$-axis.

ROC Properties (DT)

Same logic as Laplace, with circles replacing lines:

- ROC = annular ring $r_1 < |z| < r_2$.
- **Right-sided:** $|z| > \max |d_i|$ (exterior).
- **Left-sided:** $|z| < \min |d_i|$ (interior).
- DTFT exists $\iff$ unit circle $\subset$ ROC.

z-Transform Pairs

$x[n]$	$X(z)$	ROC
$\delta[n]$	1	all z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\lvert z\rvert > 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\lvert z\rvert > \lvert a\rvert$
$\delta[n-k]$	z^{-k}	$\lvert z\rvert > 0$
$r^n \cos\omega_0 n\, u[n]$	$\frac{1 - r\cos\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$
$r^n \sin\omega_0 n\, u[n]$	$\frac{r\sin\omega_0\, z^{-1}}{1 - 2r\cos\omega_0\, z^{-1} + r^2 z^{-2}}$	$\lvert z\rvert > r$

z-Transform Properties

Property	Sequence	z-Domain
Linearity	$ax_1 + bx_2$	$aX_1 + bX_2$
Time shift	$x[n-n_0]$	$z^{-n_0}X(z)$
z-scaling	$z_0^n x[n]$	$X(z/z_0)$
Convolution	$x_1 * x_2$	$X_1 X_2$
Diff. in z	$n\,x[n]$	$-z\frac{d}{dz}X(z)$

Inverse z-Transform

Same strategy as Laplace but in z^{-1}:

1. Make $X(z)$ proper in z^{-1}; long-divide if needed.
2. Partial fractions: $\frac{A_i}{1-d_i z^{-1}}$.
3. ROC outside $|d_i| \to d_i^n u[n]$; inside $\to -d_i^n u[-n-1]$.

IVT / FVT (z-domain)

IVT / FVT (z)

$$x[0] = \lim_{z\to\infty} X(z)$$
$$\lim_{n\to\infty} x[n] = \lim_{z\to 1}(1 - z^{-1})X(z)$$

FVT valid **only if** all poles of $(1 - z^{-1})X(z)$ are inside the unit circle.

DT System Analysis (Lctr 21)

Transfer Function $H(z)$

From a difference eq. $\sum a_k y[n-k] = \sum b_k x[n-k]$: replace each delay $\to z^{-k}$:

$$H(z) = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}$$

Stability (DT, Causal)

Golden Rule – DT

Causal + BIBO stable $\iff$ **all poles strictly inside the unit circle** ($|p_i| < 1\ \forall i$).

Pole location	Status
All $\lvert p_i\rvert < 1$	Stable
Any $\lvert p_i\rvert > 1$	Unstable
On unit circle, none outside	Marginally stable

Unilateral z-Transform

Sum starts at $n = 0$; initial conditions appear via the shift property:

Unilateral z Shift Property

$$y[n-1] \xrightarrow{\mathcal{Z}_u} z^{-1}Y(z) + y[-1]$$
$$y[n-2] \xrightarrow{\mathcal{Z}_u} z^{-2}Y(z) + y[-2] + y[-1]z^{-1}$$

Sign trap

Laplace uses $-y(0^-)$; the z-transform uses $+y[-1]$.

Pipeline: difference eq. $\to$ $\mathcal{Z}_u$ $\to$ solve $Y(z) \to$ PFE $\to$ invert.
Total response = ZSR + ZIR.

Sampling (Lctr 22)

Impulse-Train Sampling

Multiply $x(t)$ by $p(t) = \sum \delta(t - nT)$. In frequency:

Sampled Spectrum

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X\big(j(\omega - k\omega_s)\big), \quad \omega_s = \frac{2\pi}{T}$$

Sampling = copy-paste the spectrum at every multiple of ω_s, scaled by $1/T$.

Sampling Theorem

Shannon / Nyquist

Perfect reconstruction from $x(nT)$ requires $\omega_s > 2\omega_M$ (**strict inequality**).
$2\omega_M$ = Nyquist rate. Reconstruct with ideal LPF (gain T, cutoff $\omega_M < \omega_c < \omega_s - \omega_M$).

Key Sampling Formulas

Concept	Formula
Sampling period	$T = 2\pi/\omega_s$
Nyquist rate	$2\omega_M$ (must be *exceeded*)
Aliased frequency	$f_{\text{alias}} = \lvert f_s - f_0 \rvert$ $(f_s/2 < f_0 < f_s)$
CT↔DT freq.	$\Omega = \omega T$
Reconstruction	$x_r(t) = \sum_n x(nT)\,\text{sinc}\left(\frac{t-nT}{T}\right)$

Aliasing & Practical Notes

Aliasing is irreversible

Overlapping replicas cannot be separated. Prevent with an **anti-aliasing LPF before** the sampler.

- Squaring a signal **doubles** its bandwidth.
- $\omega_s = 2\omega_M$ is **not** sufficient (e.g. $\sin(\omega_s t/2)$ sampled at ω_s gives all-zero samples).
- ZOH / FOH are practical but **inexact** approximations of sinc interpolation.

Linear Feedback Systems (Lctr 23)

Closed-Loop Transfer Function

Feedback Formula

$$Q(s) = \frac{H(s)}{1 + G(s)\,H(s)} \quad \text{(negative feedback)}$$

Positive feedback: replace + with − in the denominator.
$1 + GH = 0 \Rightarrow$ closed-loop poles (determines stability).

Block Diagram Rules

- **Cascade** → multiply. **Parallel** → add. **Feedback** → $H/(1+GH)$.
- Strategy: simplify innermost loop first, work outward.

Stability via Closed-Loop Poles

All closed-loop poles in LHP (CT) or inside unit circle (DT) ⇒ stable.

Condition on GH	$Q \approx$	Meaning
$\lvert GH \rvert \gg 1$	$1/G$	Feedback dominates
$\lvert GH \rvert \ll 1$	H	Feedback negligible
$GH = -1$	∞	**Instability**

Nyquist Plot & Criterion

Plot $G(j\omega)H(j\omega)$ in the complex plane ($\omega : -\infty \to +\infty$).

Nyquist Criterion (simplified)

Open-loop stable ⇒ closed-loop with gain K is stable iff the Nyquist contour **does not encircle** $-1/K$.

Gain Margin & Phase Margin

Instability = $\lvert GH \rvert = 1$ (0 dB) and $\angle GH = -180°$ **simultaneously**.

GM and PM

GM: At ω_1 where $\angle GH = -180°$: $\text{GM}_{\text{dB}} = 0 - \lvert GH(\omega_1) \rvert_{\text{dB}}$
PM: At ω_2 where $\lvert GH \rvert = 0$ dB: $\text{PM} = 180° + \angle GH(\omega_2)$
Both positive ⇒ **stable**. Max delay: $\tau_{\max} = \text{PM (rad)}/\omega_2$.

Reading Bode plots

GM is read from the *magnitude* plot at the −180° frequency.
PM is read from the *phase* plot at the 0 dB frequency.
Don't mix them up!

CT vs. DT Quick Reference

	Laplace (CT)	**z (DT)**
Variable	$s = \sigma + j\omega$	$z = re^{j\omega}$
FT on	$j\omega$-axis	unit circle
ROC shape	strip	ring
Right-sided	$\sigma > \sigma_{\max}$	$\lvert z \rvert > r_{\max}$
Stable (causal)	poles in LHP	poles $\lvert p \rvert < 1$
Basic pair	$\frac{1}{s+a}$	$\frac{1}{1-az^{-1}}$
Delay	e^{-st_0}	z^{-n_0}
Unil. IC	$sY - y(0^-)$	$z^{-1}Y + y[-1]$
IVT	$\lim_{s\to\infty} sX$	$\lim_{z\to\infty} X$
FVT	$\lim_{s\to 0} sX$	$\lim_{z\to 1}(1-z^{-1})X$
Feedback	$H/(1+GH)$ (same form)	

Common Mistakes to Avoid

1. Omitting the ROC – the transform is ambiguous without it.
2. ROC direction: right-sided → right/outside; left → left/inside.
3. Repeated pole of order n needs n partial-fraction terms.
4. Complex poles: complete the square, don't split conjugates.
5. FVT requires all poles of $sX(s)$ in LHP (or $(1-z^{-1})X(z)$ inside unit circle).
6. DT stability: check $\lvert a \rvert < 1$, not $a < 1$ (pole at $z = -0.9$ is stable).
7. Unilateral sign: Laplace $-y(0^-)$; z-transform $+y[-1]$.
8. Don't mix Hz and rad/s ($\omega = 2\pi f$).
9. $\omega_s = 2\omega_M$ is **not enough**; need strict $>$.
10. Negative feedback → $1 + GH$; positive → $1 - GH$.
11. GM from magnitude plot; PM from phase plot.

Practice Questions – Lecture 22 (Sampling)

1. $X(j\omega) = 0$ for $|\omega| > 5000\pi$. Find the Nyquist rate and the max allowable T.
2. True or false: $\omega_s = 2\omega_M$ guarantees perfect reconstruction.
3. In one sentence, why does sampling in time create copies of the spectrum?
4. Overlapping replicas appear in $X_p(j\omega)$. Can a filter undo this? Why or why not?
5. What is an anti-aliasing filter? Does it go before or after the sampler?
6. Nyquist rate of $x(t) = \cos(600\pi t) + \cos(1800\pi t)$?
7. $\omega_M = 3000\pi$, $T = 2 \times 10^{-4}$ s. Compute ω_s; does aliasing occur?
8. $x(t) = \cos(2\pi \cdot 900\, t)$, $f_s = 1000\,\text{Hz}$. Reconstructed frequency?
9. Same signal, $f_s = 1500\,\text{Hz}$. Reconstructed frequency?
10. Signal has Nyquist rate ω_0. Nyquist rate of $x(t-5)$? Of $x(2t)$?
11. Why does the ideal reconstruction LPF have gain T?
12. Is ZOH reconstruction exact? What error does it introduce?
13. Camera at 24 fps, blade at 25 rev/s. What does the blade look like on film?
14. $X(j\omega) = 0$ for $|\omega| > 10{,}000\pi$; $f_s = 8000\,\text{Hz}$. Is the theorem satisfied? Min f_s?
15. $x(t)$ is not band-limited. Can we ever sample it? What do real systems do?

Practice Questions – Lecture 23 (Linear Feedback Systems)

Block Diagram Simplification

1. Cascade + unity negative feedback: $H_1 = \frac{2}{s+1}$, $H_2 = \frac{1}{s+4}$. Find $Q(s)$.
2. Parallel forward path $\frac{3}{s+2} + \frac{1}{s+5}$, feedback $G = 4$. Find $Q(s)$.
3. Nested loops: inner forward $1/s$, inner feedback 3; outer unity feedback. Find $Q(s)$.

Closed-Loop Poles & Stability

4. $H(s) = K/(s+5)$, $G = 1$.
   (a) Closed-loop TF and pole?
   (b) Stable for which K?
   (c) K for pole at $s = -10$?
5. $H(s) = K/[(s+1)(s+2)]$, $G = 1$.
   (a) Characteristic equation?
   (b) Range of K for stability?
6. DT: $H(z) = Kz^{-1}/(1 - 0.8z^{-1})$, $G = 1$.
   (a) Closed-loop pole vs. K?
   (b) $K > 0$ range for stability?

Gain & Phase Margins

7. Bode data: at $\omega_2 = 5$ ($|GH| = 0\,\text{dB}$): $\angle GH = -150°$. At $\omega_1 = 20$ ($\angle GH = -180°$): $|GH| = -8\,\text{dB}$.
   (a) Phase margin?
   (b) Gain margin?
   (c) Stable?
   (d) Max tolerable time delay?

True / False

8. Negative feedback always stabilizes a system.
9. Phase margin of 60° $\Rightarrow$ stable.
10. A Nyquist plot is $G(j\omega)H(j\omega)$ plotted in the complex plane.

Good luck on the exam!